

Q-Q Plot (Quantile-Quantile Plot)

Definition

A **Q-Q Plot (Quantile-Quantile Plot)** is a graphical technique used to determine whether a dataset follows a specific theoretical distribution, most commonly the **normal distribution**.

It compares:

- **Theoretical quantiles** (expected values from a normal distribution)
- **Sample quantiles** (actual values from your dataset)

Purpose of Q-Q Plot

The Q-Q plot helps to:

- Check data normality
- Detect skewness
- Identify outliers
- Understand tail behavior
- Decide whether data transformation is needed

Axes of a Q-Q Plot

- **X-axis:** Theoretical quantiles from a normal distribution
- **Y-axis:** Quantiles of the actual data

Each point represents a pair of corresponding quantiles.

Interpretation

1. Normal Distribution

If the points lie approximately on a straight diagonal line:

- Data is normally distributed
- Mean $\approx$ Median $\approx$ Mode
- No transformation is required

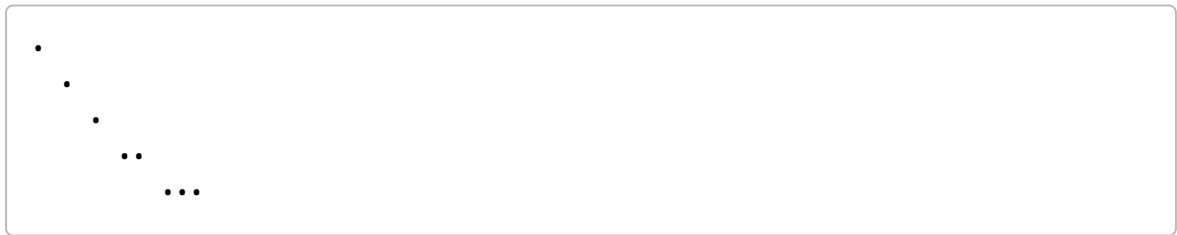




2. Right-Skewed Distribution (Positive Skew)

If the points bend upward at the upper-right end:

- Data has a long right tail
- Large values occur more frequently



Recommended transformations:

- Log transformation
- Square root transformation
- Reciprocal transformation

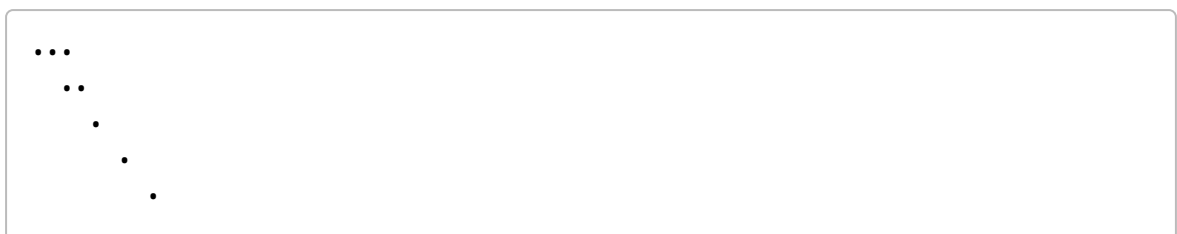
Examples:

- Income
- House prices
- Population

3. Left-Skewed Distribution (Negative Skew)

If the points bend downward at the lower-left end:

- Data has a long left tail
- Small values occur more frequently



Recommended transformations:

- Square transformation
- Cube transformation

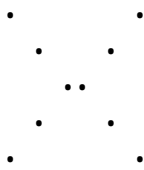
Examples:

- Easy exam scores
 - Customer ratings
-

4. Heavy Tails (Outliers Present)

If both ends move away from the straight line:

- Dataset contains extreme values
- More outliers than a normal distribution



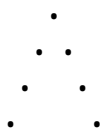
Recommended actions:

- Remove outliers
 - Use RobustScaler
 - Apply log transformation
-

5. Light Tails

If both ends bend inward toward the line:

- Fewer extreme values than normal
- Data is concentrated around the center



Usually, no transformation is needed.

Python Code

```
import scipy.stats as stats
import matplotlib.pyplot as plt

stats.probplot(df["column_name"], dist="norm", plot=plt)
```

```
plt.title("Q-Q Plot")  
plt.show()
```

When to Use Q-Q Plot

Use Q-Q plots:

- Before applying feature transformations
- Before statistical tests that assume normality
- Before using algorithms sensitive to non-normal data

Examples of such algorithms:

- Linear Regression
- Logistic Regression
- Naive Bayes
- PCA
- KNN
- SVM

Tree-based models usually do not require normal data:

- Decision Trees
- Random Forest
- XGBoost

Important Rule

Histogram → Shows data shape

Q-Q Plot → Confirms normality

Always use Q-Q plots together with:

- Histogram
- Boxplot
- Skewness value